

Release Note SDK 2.0.0 & Modem 2.0.0

Big Upgrade: LA 4.0 → LA 5.0 (Cavli CQS290 / QCM2290 IoT)

Cavli SDK Release: 2.0.0 — based on Qualcomm QCM2290.LA.5.0
Modem version: 2.0.0 (MPSS.HA.1.1-00549)
Platform: Qualcomm QCM2290 (Cavli CQS290 module)
From: Snapdragon_Mid_2020_IOT.SPF.4.0 — QCM2290.LA.4.0 (RNO-240710023155, 10-Jul-2024)
To: IOT_High_Mid_2024.SPF.1.0 — QCM2290.LA.5.0 (RNO-260220023709, 20-Feb-2026)
Android: 14 (U) — Linux kernel: 4.19 → 5.15 (GKI, Kernel Platform 2.0)

1. Overview

This is a **big upgrade** from the `Snapdragon_Mid_2020_IOT.SPF.4.0` program (internally QCM2290.LA.4.0) to the `IOT_High_Mid_2024.SPF.1.0` program (internally QCM2290.LA.5.0).

Major architectural changes:

- Kernel:** moved from the legacy `LA.UM.9.15.2` kernel (Linux 4.19) to **Kernel Platform 2.0** `KERNEL.PLATFORM.2.0.r12` (Linux 5.15 GKI) — kernel/vendor split following the GKI model.
- HLOS System:** `LA.QSSI.14.0.r1` → `LA.QSSI.14.0.r3` ; Android 11 branch (`LA.QSSI.11.0`) dropped.
- HLOS Vendor:** `LA.UM.9.15.2` → `LA.VENDOR.13.2.1.r2` (Divar vendor line), with new `LA.QISI.13.0.r1`.
- ADSP:** major bump `ADSP.VT.5.4.1` → `ADSP.VT.5.4.3.c1` (Divar).
- Synchronized updates to MPSS, TZ, BOOT, BTFM, WLAN, RPM.

This release's changes are organized into two domains:

- Part A — Android SDK (HLOS):** peripherals, functions, features, EDK2 bootloader, kernel drivers, frameworks, vendor apps. See Section 2.
- Part B — Chipcode (Firmware):** MPSS (modem), BOOT, TZ, RPM, ADSP and their build tooling. See Section 3.

The full LA 4.0 → LA 5.0 Build ID comparison table is in **Appendix A**.

2. Part A — Android SDK (HLOS): Peripherals, Functions & Features

Peripheral bring-up and feature enablement on the Android/HLOS side (bengal + qssi source trees).

Supported peripherals (CQS290 EVK):

Category	Support
Display	MIPI DSI 1280x800 LCD
Input / Touch	Goodix GT911 capacitive touch (I2C)
Audio	3.5 mm headphone jack
Multimedia / Camera	2x Sony IMX258 CSI camera (PDAF)
Connectivity	WLAN 1x1 802.11a/b/g/n/ac, Bluetooth 5.0, GNSS

Android SDK change set (Cavli commits):

Commit	Area	Change
429e15c	vendor / camera	Add IMX258 PDAF to CHI-SDK
6606492	msm-kernel / power	Charger driver: skip battery checks (battery-less module)
bb9d61f	proprietary / storage	Disable HS400 Enhanced Strobe on new HW rev (fall back to HS400)
8887e23	proprietary / power	Remove PSY_IIO (battery-less config)

Commit	Area	Change
0a524a1	vendor/cavli / OTA	Automotive support + OTA file:// / content:// URI handling
a369820	device/qcom	Change product model name
4ef3e4b	qssi/device/qcom	Change product model name (system)
fb32dfc	edk2	EDK2 bootloader aligned to Kernel Platform 2.0
3bc828b	vendor/cavli/build-tools	Enable job build configuration
ced2ca2	vendor/cavli/setup	Suppress change-log output during compile
c8d4a62	qssi/device/qcom	HLOS baseline sync — LA.QSSI.14.0.R3
5688606	qssi/frameworks	HLOS baseline sync — LA.QSSI.14.0.R3
1771cc0	qssi/vendor	HLOS baseline sync — LA.QSSI.14.0.R3
e7e29b0	bengal/device/qcom	Vendor baseline sync — LA.VENDOR r1_000020.1

2.1 Display / Touch / Audio

- **Display:** MIPI DSI 1280x800 LCD panel brought up to Android UI.
- **Touch:** Goodix GT911 capacitive touch controller over I2C.
- **Audio:** 3.5 mm headphone jack playback/record path.

2.2 Camera — IMX258 PDAF [429e15c](#)

Brought up 2x Sony IMX258 CSI image sensors with PDAF (Phase Detection Auto Focus) support, integrated into the Qualcomm CHI-SDK camera stack.

2.3 Storage — eMMC HS400 tuning [bb9d61f](#)

On the new HW revision, HS400 Enhanced Strobe fails the bus-width switch at 200 MHz (CMD6 returns -EILSEQ / -84 during devfreq scale-up), flooding CQHCI error dumps and triggering a watchdog reset. The interface now disables HS400ES and falls back to plain HS400 (CMD21 tuning), giving enough timing on the affected silicon.

2.4 Power — Battery-less operation [6606492](#) [8887e23](#)

The CQS290 module has no external Qualcomm SMB charger IC and no battery. The charger driver now skips battery current / presence / status checks (debug-battery mode) to prevent false errors, and PSY_IIO is removed from the kernel config for the battery-less configuration.

2.5 OTA Update & Applications [0a524a1](#)

- **Local OTA update application (SystemUpdater):**
- Detects the automotive platform and uses /data/local/tmp as the default update path.
- Separates file:// and content:// URI handling with proper permission management, plus a File-API based discovery path for file:// sources.
- Bundled utility / validation applications: GPS Test, Open Camera, Organic Maps, and the ATI app.

2.6 Product Identity [a369820](#) [4ef3e4b](#)

Product model name presented to the OS updated to Android IOT (vendor + system device configs).

2.7 Bootloader (EDK2) [fb32dfc](#)

Bootloader (EDK2) aligned to the new Kernel Platform 2.0 (KERNEL.PLATFORM.2.0.R12).

2.8 Build / Setup Tooling & HLOS Baseline [3bc828b](#) [ced2ca2](#) [c8d4a62](#) [5688606](#) [1771cc0](#) [e7e29b0](#)

Vendor build-tools enabled for job builds; setup suppresses change-log output during compile. HLOS baseline sync to LA.QSSI.14.0.r3 (system device/frameworks/vendor) and the LA.VENDOR.13.2.1.r2 vendor line.

3. Part B — Chipcode (Firmware): Modem, Boot, TZ, RPM, ADSP

Low-level firmware bring-up and customizations in the chipcode/ tree.

Chipcode change set (Cavli commits):

Commit	Image	Change
533743e	MPSS.HA.1.1 (Modem)	ATCoP: Cavli vendor extensions + ATI / +RFI commands, EFS identity
dcc4e79	BOOT.XF.4.1	Enable PM_LDO_17 for RF antenna path
71151f6	build-tools	Build scripts + configuration for IoT LA 5.0
d335cd1	ADSP.VT.5.4.3.c1	Baseline sync — <code>iot_high_mid_2024_spf_1_0</code> r1.0_00020.1
dd6c512	RPM.BF.1.10	Baseline sync — <code>iot_high_mid_2024_spf_1_0</code> r1.0_00020.1
34db77f	chipcode (top)	Baseline sync — <code>iot_high_mid_2024_spf_1_0</code> r1.0_00020.1

3.1 Modem (MPSS) — Vendor AT Command Interface 533743e

- **Cavli vendor AT command set** for device identification and RF information, under `interface/atcop/cavli/`:
- ATI overridden to report the full module identity: manufacturer, model, description, IMEI / SVN, part number, serial number, hardware version, modem version, and build date.
- New +RFI command reports the RFC version and QCN version.
- Device identity (HW/SW version, part number, serial number, QCN version) is persisted in the modem EFS under `/nv/item_files/cavli/` and read back via `cav_efs`.

3.2 Boot — RF Front End / Power dcc4e79

PM_LDO_17 is enabled at the bootloader (BOOT.XF.4.1) stage to supply power to the RF antenna path.

3.3 RPM / ADSP baseline dd6c512 d335cd1

RPM and ADSP synchronized to the LA 5.0 firmware baseline (`iot_high_mid_2024_spf_1_0`, r1.0_00020.1). ADSP is a major bump ADSP.VT.5.4.1 → ADSP.VT.5.4.3.c1 (Divar).

3.4 TZ — QUP / QP Access Control

QUPv3 Serial Engine (SE) allocation and TrustZone (TZ) access-control mapping used on the CQS290 module (default profile, from QUPAC_Access.c).

SE	Protocol	Mode	NS Owner	AllowFifo	Load	ModExcl	Function (Peripheral)
QUPV3_0_SE0	UART_4W	FIFO	AC_HLOS	TRUE	TRUE	FALSE	Free — available for external peripheral (UART)
QUPV3_0_SE1	I2C	FIFO	AC_HLOS	TRUE	TRUE	FALSE	SMB / EEPROM / PM8008 (PMIC)
QUPV3_0_SE2	I2C	GSI	AC_HLOS	FALSE	TRUE	FALSE	Legacy Touch
QUPV3_0_SE3	UART_4W	FIFO	AC_HLOS	TRUE	TRUE	FALSE	BT HCI
QUPV3_0_SE4	UART_2W	FIFO	AC_HLOS	FALSE	TRUE	FALSE	Debug UART
QUPV3_0_SE5	SPI	GSI	AC_HLOS	FALSE	TRUE	TRUE	Free — available for external peripheral (SPI)

Note: In the Qualcomm reference design SE0 is used for the NFC eSE (UART) and SE5 for a Fingerprint sensor (SPI). The Cavli CQS290 EVK does not populate NFC or a Fingerprint sensor, so these two Serial Engines are left free and are available for the customer to connect their own external peripherals (SE0 as UART, SE5 as SPI). The remaining SEs are assigned as shown above.

3.5 Chipcode Build System 71151f6

Dedicated LA 5.0 build scripts and configuration for ADSP, Boot, CDSP, Modem (MPSS), RPM, and TrustZone.

4. Test Results — Qualcomm LA 5.0 (Qualcomm reference board)

The results below are reproduced from Qualcomm's LA 5.0 release note (RNO-260220023709) and were obtained on Qualcomm's reference/QRD platform, not on Cavli hardware. They confirm the software baseline.

4.1 Functional / Peripheral Test Cases

Test Case	Result
Build loading (PCAT / JTAG / Fastboot)	PASS
UART shell	PASS
FBC till Android UI (Display)	PASS
ADB (shell / push / remount)	PASS
ADB reboot (bootloader / edl) & Fastboot reboot	PASS
ADSP / CDSP / MPSS / WPSS PIL	PASS
Camera front/rear (preview + snapshot)	PASS
Camcorder front/rear	PASS
Audio playback/record + AV playback/record	PASS
Graphics	PASS
Dump collection	PASS
BT connect & pairing	PASS
WiFi connect & browsing	PASS

4.2 Modem Test Cases (dual-SIM)

Combination	MO SIM-1	MO SIM-2	MT SIM-1	MT SIM-2	Data SIM-1	Data SIM-2	SMS SIM-1	SMS SIM-2
GSM + GSM	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass
GSM + UMTS	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass
GSM + LTE	Pass	NA	Pass	NA	Pass	–	Pass	NA
UMTS + UMTS	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass
UMTS + LTE	Pass	NA	Pass	NA	Pass	Pass	Pass	NA
UMTS + GSM	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass
LTE + GSM	NA	Pass	NA	Pass	Pass	Pass	NA	Pass
LTE + UMTS	NA	Pass	NA	Pass	Pass	Pass	NA	Pass

5. Known Issues

- Kernel version mapping (4.19 → 5.15 GKI) is inferred from the Qualcomm program designations and is not explicitly printed in the Qualcomm release-note PDF.
- SSC QUP per-engine sensor assignment is not named in the source table (bus/ownership only — see 3.4).

Appendix A — Build ID Comparison (QCM2290 META Build)

LA 4.0 Meta Build: QCM2290.LA.4.0-00041-STD.PROD-1 (Milestone Post-CS7) **LA 5.0 Meta Build:** QCM2290.LA.5.0-00053-STD.PROD-1 (Milestone Post-CS8)

Software Image	LA 4.0 Build ID	LA 5.0 Build ID	Note
META	QCM2290.LA.4.0-00041-STD.PROD-1	QCM2290.LA.5.0-00053-STD.PROD-1	Big upgrade 4.0 → 5.0
Kernel	LA.UM.9.15.2.r1-09900-KAMORTA.QSSI14.0-1 (kernel 4.19)	KERNEL.PLATFORM.2.0.r12-06100-kernel.0-1 (kernel 5.15 GKI)	New kernel platform
HLOS System	LA.QSSI.14.0.r1-15800-qssi.0-1	LA.QSSI.14.0.r3-03300-qssi.0-1	r1 → r3
HLOS System (A11)	LA.QSSI.11.0.r1-22500-qssi.0-1	(removed)	Android 11 dropped
HLOS Vendor	(part of LA.UM.9.15.2)	LA.VENDOR.13.2.1.r2-14000-DIVAR.QSSI14.0-1	New vendor line
QISI	–	LA.QISI.13.0.r1-02300-qssi.0-1	New in LA5.0
ADSP	ADSP.VT.5.4.1-00389.8-KAMORTA-1	ADSP.VT.5.4.3.c1-00118-DIVAR-1	Major bump
BOOT	BOOT.XF.4.1-00392-KAMORTALAZ-1	BOOT.XF.4.1-00408-KAMORTALAZ-1	00392 → 00408
MPSS (Modem)	MPSS.HA.1.1-00471-KD_ALL_PACK-1	MPSS.HA.1.1-00549-KD_ALL_PACK-2	00471 → 00549
RPM	RPM.BF.1.10-00224-KAMORTAAAAAANAZR-1	RPM.BF.1.10-00225-KAMORTAAAAAANAZR-1	00224 → 00225
TZ.APPS	TZ.APPS.2.0-00281-KAMORTAAAAAANAZT-2	TZ.APPS.2.0-00340-KAMORTAAAAAANAZT-1	00281 → 00340
TZ.XF	TZ.XF.5.1-01183-ADJKAMORTAAANAZT-2	TZ.XF.5.1-01213-ADJKAMORTAAANAZT-2	01183 → 01213
BTFM.CMC.1.2.0	BTFM.CMC.1.2.0-00218-QCACHROMZ-2	BTFM.CMC.1.2.0-00222-QCACHROMZ-3	00218 → 00222
BTFM.CMC.1.3.0	BTFM.CMC.1.3.0-00115-QCACHROMZ-1	BTFM.CMC.1.3.0-00153-QCACHROMZ-1	00115 → 00153
VIDEO	VIDEO.VE.6.0-00053-PROD-2	VIDEO.VE.6.0-00053-PROD-2	Unchanged
WLAN	WLAN.HL.3.3.7.c1-00515-QCAHLSWMTPLZ-1	WLAN.HL.3.3.7.c1-00553-QCAHLSWMTPLZ-1	00515 → 00553
WLAN_ADDON	WLAN_ADDON.HL.1.0-00078-CNSS_RMZ_WAPI-1	WLAN_ADDON.HL.1.0-00078-CNSS_RMZ_WAPI-1	Unchanged

HLOS tags (LA 5.0): Vendor AU: AU_LINUX_ANDROID_LA.VENDOR.13.2.1.R2.11.00.00.1017.140 · Kernel AU: AU_LINUX_KERNEL.PLATFORM.2.0.R12.00.00.00.019.061 · System AU: AU_LINUX_ANDROID_LA.QSSI.14.0.R3.14.00.00.1236.033 · QISI AU: AU_LINUX_ANDROID_LA.QISI.13.0.R1.11.00.00.1316.023 · LKG: Keystone UKQ1.251204.001, Android Upstream UPM1.231025.065.